UPPSALA UNIVERSITET

Department of Information Technology

FRONT SHEET FOR EXAMS	DATE: 2024-08-26
Course name 1RT485 - Introduction to Computer Control Systems	
Your exam code	

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Semester when you were first registered to the course:	Programme:
Time for submitting the exam:	Table number:

No.	Solved problems (mark with X)	Points earned	Comments from the teacher
1			
2			
3			
4			
5			
6			
7			
8			
9			
10			

$\sum$
☐ Exam with bonus points: Grade is not shown.[1]
5 ≥ ☐ 4 ≥ ☐ 3 ≥ ☐
Exam grade:

INSTRUCTIONS

Please check that you have the correct exam!

This sheet should always be submitted, even if you haven't solved any of the exam problems.

Each solution should be written on a separate paper.

Write your exam code on each paper.

Please use only *one side* of the papers and do not use a pencil with red colour.

Sort the solutions in question order, with question 1 first, before you submit them.

1 The final result (points including bonus points and grade) can be found in Ladok after certification.

Introduction to Computer Control Systems, 5 credits, 1RT485

Date: 2024-08-26

Teacher on duty: Daniel Arnström

Number of problems: 5

Allowed aid: A non-programmable calculator and mathematical handbooks

Preliminary grades:	grade 3	15 points
	grade 4	21 points
	grade 5	26 points

Some general instructions and information:

- Your solutions should be given in English.
- Write only on one side of the paper.
- Write your exam code and page number on all pages.
- Do not use a red pen.
- Use separate sheets of paper for the different problems (i.e. the numbered problems, 1–5).

With the exception of Problem 1, ***all your answers must be clearly motivated!*** *A correct but unmotivated answer will score reduced points!*

Best of luck!

Formula sheet

Laplace transform table

Table 1: Basic Laplace transforms

$f(t)$	$F(s)$	$f(t)$	$F(s)$
unit impulse — imp(t)	1	$e^{-at}\sin(bt)$	$\frac{b}{(s+a)^2+b^2}$
unit step — step(t)	$\frac{1}{s}$	$e^{-at}\cos(bt)$	$\frac{s+a}{(s+a)^2+b^2}$
unit ramp — ramp(t)	$\frac{1}{s^2}$	$\sinh(bt)$	$\frac{b}{s^2-b^2}$
t	$\frac{1}{s^2}$	$\cosh(bt)$	$\frac{s}{s^2-b^2}$
t^n	$\frac{n!}{s^{n+1}}$	$\frac{1}{2b}t\sin(bt)$	$\frac{s}{(s^2+b^2)^2}$
e^{-at}	$\frac{1}{s+a}$	$t\cos(bt)$	$\frac{s^2-b^2}{(s^2+b^2)^2}$
$\frac{1}{a}(1-e^{-at})$	$\frac{1}{s(s+a)}$	$\frac{\cos(bt)-\cos(at)}{a^2-b^2}; (a^2 \neq b^2)$	$\frac{s}{(s^2+a^2)(s^2+b^2)}$
$\frac{1}{(n-1)!}t^{n-1}e^{-at}; (n=1,2,3...)$	$\frac{1}{(s+a)^n}$	$\frac{\sin(at)+at\cos(at)}{2a}$	$\frac{s^2}{(s^2+a^2)^2}$
$\sin(bt)$	$\frac{b}{s^2+b^2}$		
$\cos(bt)$	$\frac{s}{s^2+b^2}$		

Table 2: Properties of Laplace Transforms

$\mathcal{L}[af(t)] = aF(s)$	$\mathcal{L}[tf(t)] = -\frac{dF(s)}{ds}$
$\mathcal{L}[f_1(t)+f_2(t)] = F_1(s)+F_2(s)$	$\mathcal{L}\left[t^2 f(t)\right] = \frac{d^2}{ds^2}F(s)$
$\mathcal{L}\left[\frac{d}{dt}f(t)\right] = sF(s)-f(0)$	$\mathcal{L}[t^n f(t)] = (-1)^n \frac{d^n}{ds^n}F(s), \quad n \in \mathbb{N}$
$\mathcal{L}\left[\frac{d^2}{dt^2}f(t)\right] = s^2F(s)-sf(0)-f'(0)$	$\mathcal{L}\left[f\left(\frac{t}{a}\right)\right] = aF(as)$
$\mathcal{L}\left[\int f(t)\,dt\right] = \frac{F(s)}{s}+\frac{1}{s}\left[\int f(t)\,dt\right]_{t=0}$	$\mathcal{L}\left[\int_0^t f_1(t-\tau)f(t)d\tau\right] = F_1(s)F_2(s)$
$\mathcal{L}[f(t-a)] = e^{-as}F(s)$	$\mathcal{L}[e^{-at}f(t)] = F(s+a)$

Open-loop and sensitivity functions

$$G_o(s) = G(s)F_y(s), \qquad S(s) = \frac{1}{1+G_o(s)}, \qquad T(s) = 1-S(s)$$

State-space forms and transfer function relations

- State-space form and transfer function

$$\begin{aligned} \dot{x} &= Ax+Bu \\ y &= Cx+Du \end{aligned} \quad \Rightarrow \quad \boxed{G(s) = C(sI-A)^{-1}B+D}$$

- Associated matrices

$$\mathcal{S} = \begin{bmatrix} B & AB & \cdots & A^{n-1}B \end{bmatrix} \qquad \mathcal{O} = \begin{bmatrix} C \\ CA \\ \vdots \\ CA^{n-1} \end{bmatrix}$$

- LTI system with transfer function

$$\boxed{G(s) = \frac{b_0 s^n + b_1 s^{n-1} + \cdots + b_n}{s^n + a_1 s^{n-1} + \cdots + a_n}}$$

  i) Observable canonical form

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \\ \vdots \\ \dot{x}_n \end{bmatrix} = \begin{bmatrix} -a_1 & 1 & 0 & \cdots & 0 \\ -a_2 & 0 & 1 & \cdots & 0 \\ -a_3 & 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ -a_n & 0 & 0 & \cdots & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{bmatrix} + \begin{bmatrix} b_1 - a_1 b_0 \\ b_2 - a_2 b_0 \\ b_3 - a_3 b_0 \\ \vdots \\ b_n - a_n b_0 \end{bmatrix} u$$

$$y = \begin{bmatrix} 1 & 0 & \cdots & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} + b_0 u$$

  ii) Controllable canonical form

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \\ \vdots \\ \dot{x}_n \end{bmatrix} = \begin{bmatrix} -a_1 & -a_2 & -a_3 & \cdots & -a_n \\ 1 & 0 & 0 & \cdots & 0 \\ 0 & 1 & 0 & \cdots & 0 \\ \vdots & 0 & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{bmatrix} + \begin{bmatrix} 1 \\ 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix} u$$

$$y = \begin{bmatrix} b_1 - a_1 b_0 & b_2 - a_2 b_0 & \cdots & b_n - a_n b_0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} + b_0 u$$

- Solution to state-space equation

$$\dot{x}(t) = Ax(t) + Bu(t), \quad x(0) = x_0$$

  can be written as

$$\boxed{x(t) = e^{At} x_0 + \int_0^t e^{A\tau} B u(t-\tau) d\tau}$$

- Observer system

$$\boxed{\dot{\hat{x}} = A\hat{x} + Bu + K(y - C\hat{x})}$$

Feedback control structures

General linear feedback in Laplace form:

$$\boxed{U(s) = F_r(s)R(s) - F_y(s)Y(s)}$$

Common control structures in this form.

- PID controller:

$$F_y(s) = F_r(s) = F(s) = K_p + \frac{K_i}{s} + K_d s,$$

  where $K_p, K_i, K_d \geq 0$

- State-feedback controller with observer:

$$\begin{aligned} F_r(s) &= \left(1 - L(sI - A + KC + BL)^{-1}B\right)\ell_0 \\ F_y(s) &= L(sI - A + KC + BL)^{-1}K \end{aligned}$$

Bode's phase diagram

Consider the loop gain $G_o(s) = \mu \frac{\prod_{k=1}^{n_z}(1-\tau_k s)}{s^g \prod_{k=1}^{n_p}(1-T_k s)}$, where for simplicity $\mu > 0$. The phase of $G(i\omega)$ at a given frequency is

$$\arg\left(G_o(i\omega)\right) = -g\,90^\circ + \left[\sum_{k=1}^{n_z} \arctan\left(\tau_k \omega\right) - \sum_{k=1}^{n_p} \arctan\left(T_k \omega\right)\right] \frac{180^\circ}{\pi}$$

Routh–Hurwitz stability criterion

Let $\Phi(s) = a_0 s^n + b_0 s^{n-1} + a_1 s^{n-2} + ...$ be the characteristic polynomial.

$$\begin{array}{ccc} a_0 & a_1 & ... \\ b_0 & b_1 & ... \\ \hline c_0 & c_1 & ... \\ \vdots & \ddots & \end{array}$$

A generic element of the third row, c_k, can be computed as

$$c_k = \frac{b_0 a_{k+1} - a_0 b_{k+1}}{b_0}$$

Discrete-time state-space forms

A continuous time system with zero-order-hold input signal and sample period T can be written in discrete-time as:

$$\boxed{\begin{aligned} x_k &= Fx_k + Gu_k \\ y_k &= Hx_k \end{aligned}}$$

where

$$
\begin{aligned}
F &= e^{AT} \\
G &= \int_{\tau=0}^{T} e^{A\tau} d\tau B = \lceil \text{if } A^{-1} \text{ exists} \rceil = A^{-1}(e^{AT} - I)B \\
H &= C
\end{aligned}
$$

Discrete-time state-space forms

A continuous time system with zero-order-hold input signal and sample period T can be written in discrete-time as:

$$
\boxed{\begin{aligned} x(k+1) &= Fx(k) + Gu(k) \\ y(k) &= Hx(k) \end{aligned}}
$$

where

$$
\begin{aligned}
F &= e^{AT} \\
G &= \int_{\tau=0}^{T} e^{A\tau} d\tau B = \lceil \text{if } A^{-1} \text{ exists} \rceil = A^{-1}(e^{AT} - I)B \\
H &= C
\end{aligned}
$$

Bilinear transform formula ($\alpha \in \{0, \frac{1}{2}, 1\}$)

$$
s = \frac{1}{T} \frac{1 - q^{-1}}{\alpha + (1 - \alpha)q^{-1}}
$$

Problem 1: basic questions (6/30)

Answer only *'true'* or *'false'*. Each correct answer gives 1 point, each wrong answer gives -1 point. Minimum total points for Part A and B is 0 , respectively.

Part A

Note: Write 'skip' if your total home assignment score ≥ 8

i) The system

$$G(s) = \frac{s+1}{(s+2)(s+5)}$$

is input-output stable.

ii) The static gain of a system can be used to determine the steady-state error for step inputs.

iii) In PID controllers, the integral part allows to reduce oscillations and makes the closed-loop system faster.

(3 p)

Part B

Note: Write 'skip' if your total home assignment score ≥ 13

i) A large bandwidth for the closed-loop system means a faster system.

ii) For a system on state-space form $\dot{x} = Ax + Bu$, all eigenvalues of A need to be strictly in the **right** half-plane for the system to be asymptotically stable.

iii) A transfer function $G(s)$ can admit several minimal state-space realization.

(3 p)

Proposed solution to problem 1

Part A

i) True

ii) True

iii) False

Part B

i) True

ii) False

iii) True

Problem 2 (6/30)

Consider a system with output y and input u. Our goal is for y to follow a reference signal r. The system's transfer function from u to y is

$$G(s) = \frac{1}{s^2 + 1},$$

that is, $Y(s) = G(s)U(s)$.
Assume that we use PD-controller of the form

$$u = -\left(K \cdot e + 2\dot{e}\right),$$

where $e = y - r$.

a) What is the closed-loop transfer function (in other words, the transfer function from r to y)?

(2 p)

b) For which values of K is the closed-loop system stable?

(2 p)

c) What is the steady-state error when the reference is $r(t) = 1 \cdot \text{step}(t)$?
Note: your answer may be a function of K.

(2 p)

Proposed solution to problem 2

a) The open-loop transfer function is $G_o(s) = \frac{2s+K}{s^2+1}$, while the closed-loop transfer function is $G_c(s) = \frac{G_o(s)}{1+G_o(s)} = \frac{2s+K}{s^2+2s+1+K}$.

b) First, note that we can rewrite the denominator of $G_c(s)$ as $(s+1)^2 + K$. For $K \geq 0$, we get poles in $-1 \pm i\sqrt{K}$. For $K \leq 0$ we get poles in $-1 \pm \sqrt{|K|}$, which have a strictly negative real-part as long as $|K| < 1$. Taken together, we have that $K > -1$ yields a stable system.

c) When the reference is a unit step ($R(s) = \frac{1}{s}$), the closed-loop output is $Y(s) = G_c(s)R(s)$. Then, according to the Final Value theorem,

$$\begin{aligned} e(\infty) &= \lim_{s\to 0} s\left(Y(s) - R(s)\right) = \lim_{s\to 0} s(G_c(s) - 1)\frac{1}{s} \\ &= \lim_{s\to 0} \left(G_c(0) - 1\right)\frac{s}{s} = \frac{K}{K+1} - 1 = -\frac{1}{K+1} \end{aligned}$$

Note, however, that this is only valid when the closed-loop system is stable (i.e., when $K > -1$)

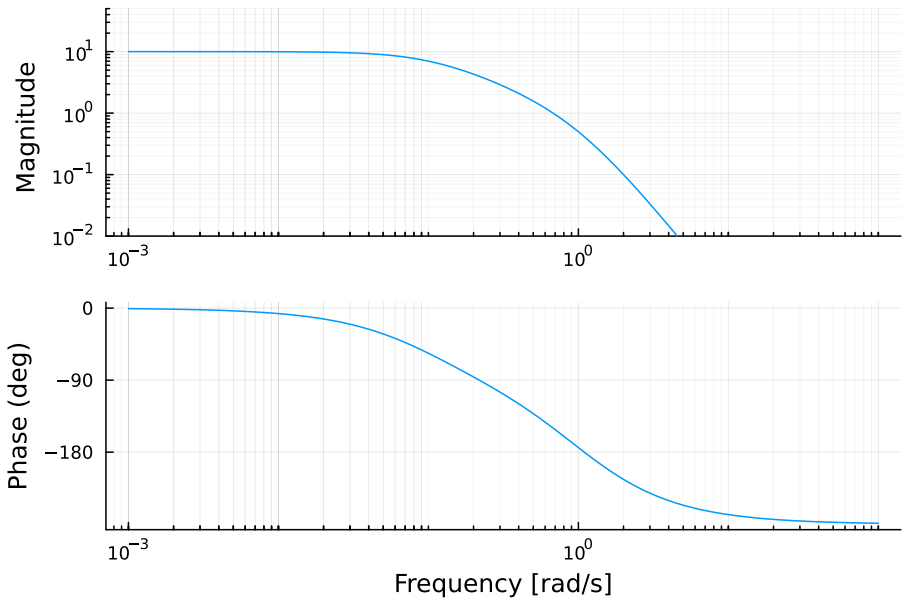

Figure 1: Bode plot for G.

Problem 3 (6/30)

Consider the system G, characterized by the bode plot given in Figure 1. (Note that the magnitude is given in absolute scale rather than dB.)

a) What is the crossover frequency and phase margin for the system? Note: only approximate values are required.

(2 p)

b) We are now interested in using a P-controller with gain K to get the crossover frequency at $\omega_c = 2$ rad/s. What value of K yields the crossover frequency $\omega_c = 2$ rad/s?

(2 p)

c) If you use the P-controller designed in b), what will be the new phase margin? Based on this, what will be the properties of the closed-loop system?

(2 p)

Proposed solution to problem 3

a) The crossover frequency is the frequncy for which the magnitude is 1, which by observation is ≈ 0.7 rad/s. The phase margin is the difference between the phase at the crossover frequency and $-180°$, which by observation is $\approx 30°$.

b) From the Bode plot we have that the current gain at $\omega = 2$ rad/s is $|G(2i)| \approx 0.1$. With a P-controller, the new open-loop is $K \cdot G(s)$. To make 2 rad/s the new crossover frequency, hence, requires $K|G(2i)| = 1 \implies K \approx \frac{1}{0.1} = 10$.

c) Since a P-controller will not the affect the phase plot, we can read of the new phase margin from Figure 1 as the distance between $-180°$ and the phase at $\omega = 2$ rad/s, which by observation is $\approx -30°$. A negative phase margin in this instance implies that the resulting closed-loop system will be unstable.

Problem 4 (6/30)

Consider a system with output y and input u that is described by the differential equation

$$\dot{y} - y = u.$$

a) Put the system on state-space form with the states $x_1 = y$ and $x_2 = \dot{y}$.

(1 p)

b) Is the state-space model derived in (a) controllable?

(1 p)

c) For the state-space model in (a), determine a state feedback law of the form $u = -l_1 x_1 - l_2 x_2$ such that the closed-loop poles are placed in -1 and -2.

(2 p)

d) Could a PD-controller of the form $u = -K_p y - K_d \dot{y}$ be used to stabilize the system? If yes, what are some values of K_p and K_d that yields a closed-loop system?

(2 p)

Proposed solution to problem 4

a) We have that

$$\begin{aligned}\dot{x}_1 &= x_2\\ \dot{x}_2 &= x_1 + u,\end{aligned}$$

which is of the form $\dot{x} = Ax + Bu$ with

$$A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

b) The controlability matrix is

$$\mathcal{S} = \begin{bmatrix} B & AB \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \implies \det(\mathcal{S}) \neq 0.$$

Hence, the system is controllable.

c) With $L = \begin{bmatrix} l_1 & l_2 \end{bmatrix}$, the characteristic polynomial of the closed loop system is

$$\det(\lambda I - (A + BL)) = \det \begin{bmatrix} \lambda & -1 \\ -1 - l_1 & \lambda - l_2 \end{bmatrix} = \lambda^2 - l_2\lambda - (1 + l_1) \qquad (1)$$

The desired characteristic polynomial is $(\lambda+1)(\lambda+2) = \lambda^2 + 3\lambda + 2$. Matching the coefficients of the polynomials gives

$$\begin{cases} -l_2 = 3 \\ -(1 + l_1) = 2 \end{cases} \implies \quad L = \begin{bmatrix} -3 & -3 \end{bmatrix}$$

d) Yes. In fact, the controller that was computed in c) is a PD-controller of this form! So using, for example, the values l_1 and l_2 computed in c) and setting $K_p = l_1$ and $K_d = l_2$ gives a stable closed-loop system.

Problem 5 (6/30)

We want to construct a closed-loop system with bandwidth $\omega_b = 4$ rad/s. To do this, we use a controller of the form $U(s) = F(s)E(s)$, with $F(s) = K_p + \frac{K_i}{s}$ (as usual, u and e denote the control signal and control error, respectively.)

a) What is a suitable sampling frequency for the system? What are the drawbacks of setting the sampling frequency too low or too high?

(2 p)

b) Suppose that we settle for a sampling time of $T_s = 2$ second. Discretize $F(s)$ using this sampling time and Tustin's method (in other words, with a bilinear transformation with $\alpha = 0.5$.) Answer with a transfer function in discrete time.

(2 p)

c) Given the discretized transfer function derived in b), state the explicit control law; in other words, u_k as a function of past control actions $(u_{k-1}, u_{k-2}, \dots)$ and control errors $(e_k, e_{k-1}, \dots)$. How many variables do you need to store in memory when implementing the control law in code?

(1 p)

d) Assume that the crossover frequency of the open-loop system is the same as the desired bandwidth ($\omega_c = 4$ rad/s.) If we implement the discretized controller using zero-order hold, what is the approximate phase margin loss due to the discretization?

(1 p)

Proposed solution to problem 5

a) To avoid aliasing, we want the sampling frequency ω_s to be at least twice as large as ω_b. Hence we want $\omega_s > 8$ rad/s. A too high sampling frequency might require more expensive hardware, and it might also reduce the lifespan of the actuator.

b) Tustin with $T_s = 2$ formula gives $s \leftarrow \frac{1-q^{-1}}{1+q^{-1}}$. Using this for $F(s) = K_p + \frac{1}{s}K_i$ gives

$$\begin{aligned} F_d(q) &= K_p + \frac{K_i}{\frac{1-q^{-1}}{1+q^{-1}}} \\ &= K_p + K_i \frac{1+q^{-1}}{1-q^{-1}} \\ &= \frac{K_p(1-q^{-1}) + K_i(1+q^{-1})}{1-q^1} \\ &= \frac{K_p + K_i + (K_i - K_p)q^{-1}}{1-q^{-1}} \end{aligned}$$

c) We have that $U(q) = F_d(q)E(q)$. By using F_d from (b) we get

$$(1-q^{-1})U(q) = (K_p + K_i + (K_i - K_p)q^{-1})E(q),$$

which corresponds to the difference equation

$$u_k - u_{k-1} = (K_p + K_i)e_k + (K_i - K_p)e_{k-1},$$

which yields the control law

$$u_k = u_{k-1} + (K_p + K_i)e_k + (K_i - K_p)e_{k-1}.$$

To realize this control law, we need to store the previous control signal u_{k-1} and (if $K_i \neq K_p$) the previous control error e_{k-1}.

d) The approximate phase margin loss is

$$\phi_m \approx -\frac{T}{2}\omega_c \frac{180}{\pi} = -\frac{2}{2}4\frac{180}{\pi} - \approx -229^\circ \tag{2}$$